Chunk Size Dominates Overlap: A Controlled Retrieval Experiment on a Philippine Consumer-Remedy Corpus

Mark Emmanuel P. Sagarino
College of Computer Studies
Jose Rizal University
Mandaluyong City, Philippines
markemmanuel.sagarino@my.jru.edu

***Abstract*—Twelve chunking configurations were compared on a seventeen-page corpus of Philippine consumer-remedy law using a fixed FAISS retriever, a fixed embedding model and nine questions with nineteen pre-registered evidence units. Chunk size, not overlap, governs retrieval quality on this corpus: mean Precision@3 rises from 0.154 at sizes of 500 characters or below to 0.296 at 1,000 or above, and Recall@3 from 0.356 to 0.597. The effect is categorical on the hardest questions. Every configuration below 1,000 characters scores exactly zero on all three metrics for the difficult band, because a question needing three passages cannot be served by three fragments. Zero-overlap settings were eliminated before retrieval: at three of four such settings some evidence was cut at a chunk boundary and no chunk contained it whole. The recommended configuration is 1,500 characters with a 300-character overlap, which attains the best MRR@3 of 0.667, the equal-best Recall@3 of 0.611, and does so with sixty chunks — fewer than the baseline it replaces.**

Keywords—retrieval-augmented generation; chunking; FAISS; Precision@3; Recall@3; mean reciprocal rank; evidence coverage.

I. Dataset

The corpus is a single seventeen-page PDF consolidating the four Philippine statutes that govern remedies for a defective consumer purchase — Republic Act No. 7394, the Consumer Act [1]; No. 8792, the Electronic Commerce Act [2]; No. 10642, the Philippine Lemon Law [3]; and No. 11967, the Internet Transactions Act [4] — together with the Department of Trade and Industry's published consumer guidance [5]. It was assembled and verified for Part 1 of this exercise and is reused here unchanged.

All seventeen pages return extractable text under pypdf. The pages carry 60,860 characters in total, a mean of 3,580 per page, with the shortest at 409 and the longest at 5,107. The variation matters: the chunker splits each page independently, so a short page yields a short final chunk regardless of the configured size.

Nine test questions were used, three at each difficulty level. Three were carried over from Part 1 and six were written for this experiment, each verified against the corpus before any measurement was taken. Easy questions are answered by one explicit fact in a single passage. Average questions need interpretation or two closely related facts. Difficult questions need several passages, and in two cases the answer turns on an exclusion in a statute the question does not name. A tenth item with no supporting evidence in the corpus was used as a guardrail check and excluded from the metrics, which are undefined when no relevant chunk exists.

II. Experimental Methodology

A. Configurations Tested

Twelve configurations were run. Five are prescribed by the exercise; the remaining seven were chosen to form a grid rather than a scattering of values, so that size and overlap can be separated. Chunk sizes of 300, 500, 800, 1,000, 1,500 and 2,000 characters were tested, with overlaps of 0, 10 and 20 per cent. The 0 per cent column spans four sizes and the 20 per cent column spans six, which permits the comparison of sizes at constant overlap ratio; the 300 and 500 rows carry all three ratios, which permits the comparison of overlap at constant size.

B. Controls

Everything other than chunk size and overlap was held constant. The same PDF was parsed once and its page texts reused by every configuration, so extraction variability cannot explain a difference in results. The same embedding model, all-MiniLM-L6-v2 producing 384-dimensional vectors, was loaded once and reused. Retrieval was FAISS IndexFlatIP over L2-normalised vectors, which is exact rather than approximate, so approximation is excluded as a source of variation. Retrieval depth was fixed at three, the depth at which the metrics are defined. A fresh index was built inside the loop for every configuration and its vector count asserted against the chunk count, so a stale or duplicated index halts the run rather than producing quietly wrong numbers.

C. Relevance Criteria

Ground truth was fixed before the experiment as nineteen evidence units across the nine questions. Because chunk identifiers change entirely when the chunk size changes, ground truth cannot be expressed as a list of chunk identifiers; each unit is therefore a short verbatim string drawn from the corpus — the minimum span that carries the fact. A retrieved chunk covers a unit when the chunk text contains that string. The test is mechanical and identical at every configuration.

The exercise specifies that mentioning the same topic or keyword is not sufficient for relevance. Anchoring on a decisive phrase enforces this: the unit for the four-repair rule is "FOUR (4) SEPARATE REPAIR ATTEMPTS", not the word "repair". Anchors were deliberately kept short, because a long anchor would fail at small chunk sizes merely by being cut, measuring the anchor rather than the chunking.

D. Metrics

Precision@3 is the proportion of the three returned chunks carrying at least one evidence unit, divided by three rather than by the number returned, so a configuration is not rewarded for returning fewer chunks. Recall@3 is the

proportion of a question's evidence units covered by those three chunks, with a chunk containing several units counting toward each. Reciprocal rank is the inverse rank of the first relevant chunk, and the overall MRR@3 is the mean across all nine questions rather than the mean of the three difficulty-level values, as the exercise requires. The three metric functions were tested against five hand-worked cases inside the notebook before the experiment was allowed to run.

III. Results

A. Evidence Reachability

A reachability check was run before retrieval, asking of each configuration whether any chunk contains each evidence unit whole. The answer separates two failures that the metrics cannot distinguish: a retriever that ranked the right chunk too low, and a chunking that destroyed the evidence. Three configurations fail this check, and all three have zero overlap.

TABLE I. Evidence Units Lost to Chunk Boundaries

Config	Setting	Units lost	Units affected
C1	300 / 0	3 of 19	A2.U2, A3.U1, D1.U3
C4	500 / 0	2 of 19	A3.U1, D1.U3
C7	800 / 0	2 of 19	A3.U2, D2.U1
C9	1000 / 0	0 of 19	none

a. All other configurations retain all nineteen units. C9 retains them by boundary luck, not by design; it is the only zero-overlap setting that does.

This is a chunking failure, not a retrieval failure. At C1 the maximum attainable Recall@3 is 0.842 before a single query is issued, because three of the nineteen units exist in no chunk. No retriever and no prompt can recover them.

B. Configuration-Level Results

TABLE II. Retrieval Performance by Configuration

Cfg	Size	Ovl	Chunks	P@3	R@3	MRR@3
C1	300	0	210	0.148	0.333	0.389
C2	300	30	234	0.148	0.333	0.389
C3	300	60	261	0.148	0.361	0.333
C4	500	0	129	0.148	0.333	0.444
C5	500	50	143	0.185	0.444	0.500
C6	500	100	161	0.148	0.333	0.444
C7	800	0	85	0.222	0.417	0.444
C8	800	160	102	0.259	0.556	0.556
C9	1000	0	69	0.259	0.556	0.611
C10	1000	200	85	0.296	0.611	0.481
C11	**1500**	**300**	**60**	**0.296**	**0.611**	**0.667**
C12	2000	400	48	0.333	0.611	0.556

b. P@3, R@3 and MRR@3 are Precision@3, Recall@3 and mean reciprocal rank at 3. Size and Ovl are in characters. C10 is the Part 1 baseline; C11, in bold, is the recommended configuration.

TABLE III. Indexing Cost by Configuration

Cfg	Chunks	Embed (s)	Index (KB)	Query (ms)
C1	210	7.88	315.0	27.96
C2	234	7.85	351.0	256.90
C3	261	12.81	391.5	39.86
C4	129	7.54	193.5	28.56
C5	143	8.52	214.5	28.11
C6	161	7.84	241.5	37.79
C7	85	7.70	127.5	27.05
C8	102	9.35	153.0	26.73
C9	69	7.25	103.5	28.59
C10	85	19.50	127.5	50.21
C11	**60**	**13.22**	**90.0**	**44.00**
C12	48	12.43	72.0	45.64

c. The 256.90 ms figure for C2 is an outlier against a background of 27 to 50 ms and is treated as measurement noise on a shared virtual machine, not as a property of the configuration.

Precision@3 rises from 0.148 to 0.333 as chunk size grows, Recall@3 from 0.333 to 0.611, and the index shrinks from 315 KB to 72 KB. The usual trade-off between retrieval quality and storage does not appear on this corpus: the better configurations are also the cheaper ones.

C. Performance by Difficulty

The difficulty breakdown is where the result becomes categorical rather than gradual.

TABLE IV. Recall@3 and MRR@3 by Difficulty Level

Config	Easy R@3	Average R@3	Difficult R@3	Difficult MRR@3
C1 to C7	0.667	0.333 to 0.583	0.000	0.000
C8 800/160	0.667	1.000	0.000	0.000
C9 1000/0	0.667	0.833	0.167	0.167
C10 1000/200	0.667	1.000	0.167	0.111
C11 1500/300	**0.667**	**0.833**	**0.333**	**0.333**
C12 2000/400	0.667	0.833	0.333	0.167

d. Easy recall is identical at 0.667 across all twelve configurations and therefore carries no information about which configuration is better.

Every configuration with a chunk size of 800 characters or less scores exactly 0.000 on all three metrics for the difficult band. Not merely low — zero. The difficult questions require two or three evidence units drawn from different pages, and three small chunks cannot hold them. Only at 1,000 characters does anything register, and only at 1,500 does Recall@3 reach 0.333.

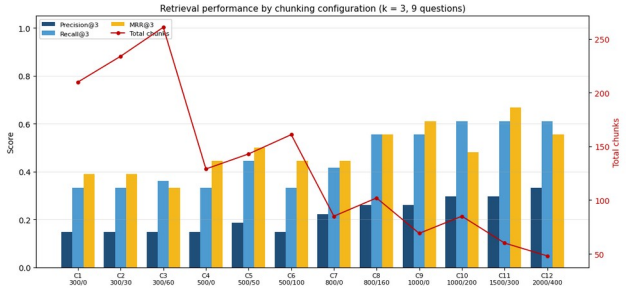


Fig. 1. Retrieval metrics by configuration, with total chunks on the right axis. Quality rises and cost falls together toward the right of the grid.

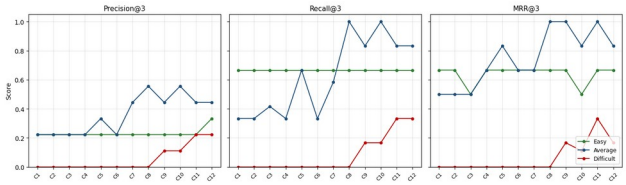


Fig. 2. The same comparison split by difficulty. The easy band is flat; the difficult band is zero until 1,000 characters.

IV. Discussion

A. Chunk Size Dominates Overlap

Aggregating across the grid, configurations of 500 characters or less attain a mean Precision@3 of 0.154, Recall@3 of 0.356 and MRR@3 of 0.417. Configurations of 1,000 or more attain 0.296, 0.597 and 0.579. Precision and recall nearly double. Overlap produces nothing comparable: within the 300-character row, moving from no overlap to 20 per cent leaves Precision@3 unchanged at 0.148 and moves Recall@3 from 0.333 to 0.361 while adding fifty-one chunks.

The reason is visible in the corpus itself. This is statutory text, in which a rule, its condition and its exception are written as consecutive sentences. A 300-character window severs them. The retriever then ranks a fragment that mentions the topic above a chunk that answers the question, because the fragment is shorter and its embedding is dominated by the matching words.

B. Overlap Earns Its Cost by Reachability, Not by Ranking

Overlap did not reliably improve ranking. At 1,000 characters it raised Recall@3 from 0.556 to 0.611 but lowered MRR@3 from 0.611 to 0.481. At 500 characters the 10 per cent setting outperformed both 0 and 20 per cent, which is not a monotone effect and is most likely an artefact of where the boundaries happen to fall in this document.

Its real contribution is categorical. Without overlap, three of four configurations lose evidence to boundaries outright. That is the argument for a non-zero overlap on this corpus, and it is an argument about coverage rather than about ranking.

C. Examples from the Retrieval Log

A successful retrieval: question A1 asks what an online buyer must do before filing a complaint and when that step is exhausted. Its two evidence units sit on pages 13 and 14, because Section 24 of the Internet Transactions Act straddles a page break and the chunker restarts on every page. No single chunk can contain both. At C8 and C10 the retriever returns one chunk from each page and Recall@3 reaches 1.000 — the two halves recovered separately and combined only in the prompt.

An unsuccessful retrieval: question D1 asks whether the Lemon Law covers a brand-new motorcycle. Answering it requires the exclusion of motorcycles on page 9, the coverage provision on the same page, and Article 100 of the Consumer Act on page 4. At every configuration below 1,000 characters the top three results are all drawn from page 10, the four-repair rule, because that passage is the closest match to the words of the question. The words that would settle the matter — the exclusion — are precisely the words the question does not use. At C11 the exclusion enters the top three and Recall@3 for the difficult band rises to 0.333.

D. The Precision Ceiling

Precision@3 never exceeds 0.333 anywhere in the grid, and this should not be read as poor performance. Most questions in this corpus are served by one or two chunks while three are always returned, so the attainable maximum is near one third for a single-chunk question. At C12 nine of twenty-seven retrieved chunks were relevant, which is the ceiling for a test set in which most questions have one containing chunk. Precision@3 here is better read as a measure of how often the retriever wastes its three slots than as an accuracy score.

E. The Embedding Window Qualifies the Large-Chunk Result

One property of the embedding model bears directly on the headline finding and is reported here rather than left implicit. The model used, all-MiniLM-L6-v2, accepts a maximum of 256 word-pieces, which corresponds to roughly a thousand characters of English prose. At C11 and C12 the chunks exceed that limit, so the vector stored in the index represents only the leading part of the chunk, while the whole chunk is returned when that vector is matched.

Part of the advantage held by the largest configurations is therefore an advantage of payload rather than of representation: the retrieved text grows while the text actually compared against the query does not. This does not overturn the two principal results. The categorical failure of every configuration at 800 characters or below is unaffected, because those chunks sit entirely inside the embedding window. The reachability result is independent of embeddings altogether, since it is a property of where the boundaries fall. But it does mean that the margin between C11 and C10 should be read as a property of this model and this window, and that a model with a longer context might place the optimum elsewhere.

F. Limitations

Four limitations qualify these results. The test set is nine questions, which is the minimum the exercise requires and is small enough that a single question changes an overall metric by about 0.11. Each configuration was run once, so the timings are single observations on a shared virtual machine and the 256.90 ms outlier at C2 illustrates the point. Only one embedding model was tested, and a model with a longer effective context might change the ranking of the larger configurations. Finally, the easy band is saturated at 0.667 recall across all twelve configurations and contributes nothing to the comparison, so the overall figures are driven by the average and difficult bands. The embedding window discussed above is a fifth qualification, and the one most likely to change the ranking of the two largest configurations if a different model were used.

V. Final Recommendation

The recommended configuration is a chunk size of 1,500 characters with an overlap of 300 characters, which is C11 in the grid.

TABLE V. Recommended Configuration Against the Baseline

Measure	C10 baseline	C11 recommended	Change
Chunk size / overlap	1000 / 200	1500 / 300	larger
Precision@3	0.296	0.296	no change
Recall@3	0.611	0.611	no change
MRR@3	0.481	0.667	+0.186
Difficult-band mean	0.130	0.296	+0.166
Total chunks	85	60	-29%
Index size	127.5 KB	90.0 KB	-29%

The selection was made in three stages. Configurations losing evidence to chunk boundaries were eliminated first, which removed C1, C4 and C7 regardless of their scores. Remaining configurations were ranked on a quality score weighting the overall mean of the three metrics at 0.6 and the difficult-band mean at 0.4, because the difficult band is the only one that separates configurations. C11 scored 0.433, ahead of C12 at 0.396 and the baseline C10 at 0.330, and no other configuration came within the 0.02 margin treated as comparable. Efficiency was therefore not needed as a tie-break, although C11 would have won it as well.

C11 is preferred to C12 despite C12's higher Precision@3 because the two have identical Recall@3 while C11 ranks the first relevant chunk substantially earlier, 0.667 against 0.556, and holds a clear advantage on difficult-band MRR, 0.333 against 0.167. In a retrieval-augmented system the rank of the first relevant chunk determines what survives a limited context budget, so an equal-recall configuration that ranks evidence higher is the better choice.

The experiment also answers a question left open by Part 1. The baseline then submitted, 1,000 characters with a 200-character overlap, is not the best configuration for this corpus. It matches C11 on precision and recall but ranks the first relevant chunk later, and it does so while storing 42 per cent more chunks. The Part 1 diagnostic had measured the motorcycle exclusion at rank 4, one place outside the scored depth, and predicted that a different chunking might lift it in. It did.

VI. Reproducibility

The submitted notebook runs the experiment end to end from the corpus PDF. It writes the configuration results, the difficulty breakdown, the 324-row retrieval log and the reachability table to a four-worksheet workbook, exports the same tables as CSV, saves both figures, and writes a JSON record of the dataset, the model, the configurations and the recommendation. Every number in this report is taken from those outputs.

The experiment is deterministic given the same corpus and embedding model: retrieval is exact rather than approximate, no sampling is involved, and the ground truth is a fixed list of verbatim strings. The only values expected to vary between runs are the timings.

TABLE VI. Final Configuration

Setting	Value
Dataset	Sagarino_PT-F1_Corpus.pdf, 17 pages, 60,860 characters
Chunk size	1,500 characters
Chunk overlap	300 characters (20 per cent)
Resulting chunks	60
Embedding model	all-MiniLM-L6-v2, 384 dimensions, L2-normalised
Vector store	FAISS IndexFlatIP (exact; cosine on unit vectors)
Retrieval depth	k = 3
Test set	9 questions, 3 per difficulty level, 19 evidence units
Precision@3 / Recall@3 / MRR@3	0.296 / 0.611 / 0.667

References

[1] Republic of the Philippines, Republic Act No. 7394, "The Consumer Act of the Philippines," Apr. 13, 1992.

[2] Republic of the Philippines, Republic Act No. 8792, "Electronic Commerce Act of 2000," Jun. 14, 2000.

[3] Republic of the Philippines, Republic Act No. 10642, "Philippine Lemon Law," Jul. 15, 2014.

[4] Republic of the Philippines, Republic Act No. 11967, "Internet Transactions Act of 2023," Dec. 5, 2023.

[5] Department of Trade and Industry, "FAQs," DTI E-Commerce Philippines, and Fair Trade Enforcement Bureau, "How to file a consumer complaint?" [Online]. Available: ecommerce.dti.gov.ph, fairtrade.dti.gov.ph

[6] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP-IJCNLP, 2019.

[7] J. Johnson, M. Douze and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Trans. Big Data, vol. 7, no. 3, pp. 535-547, 2021.

[8] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. NeurIPS, 2020.